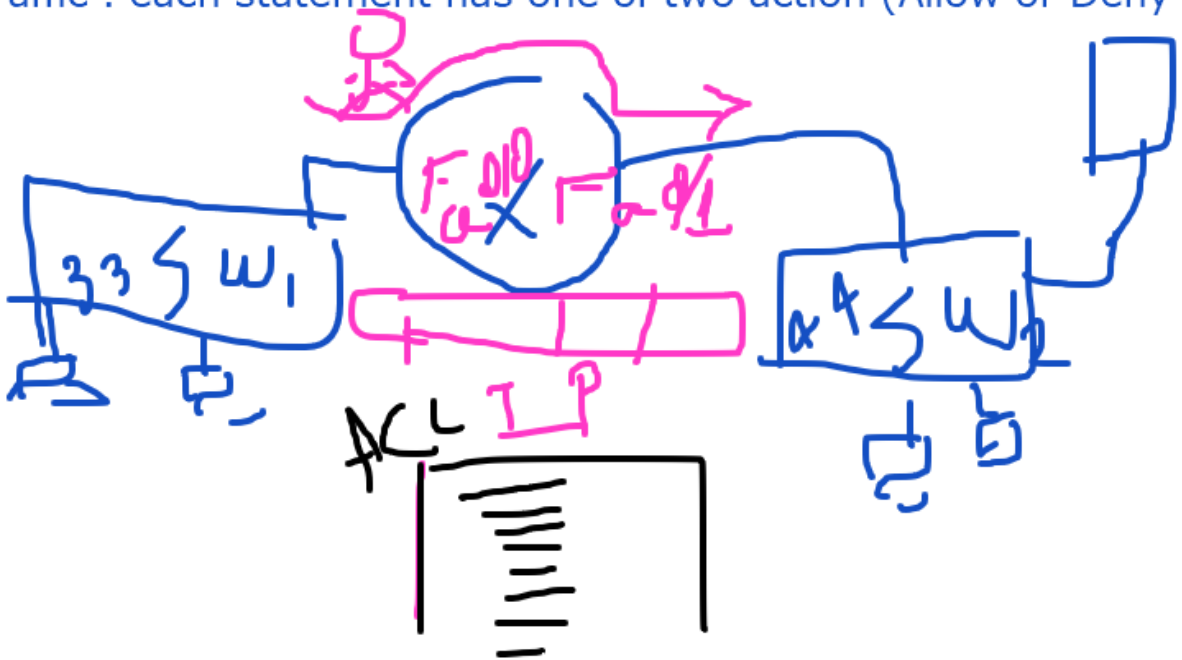
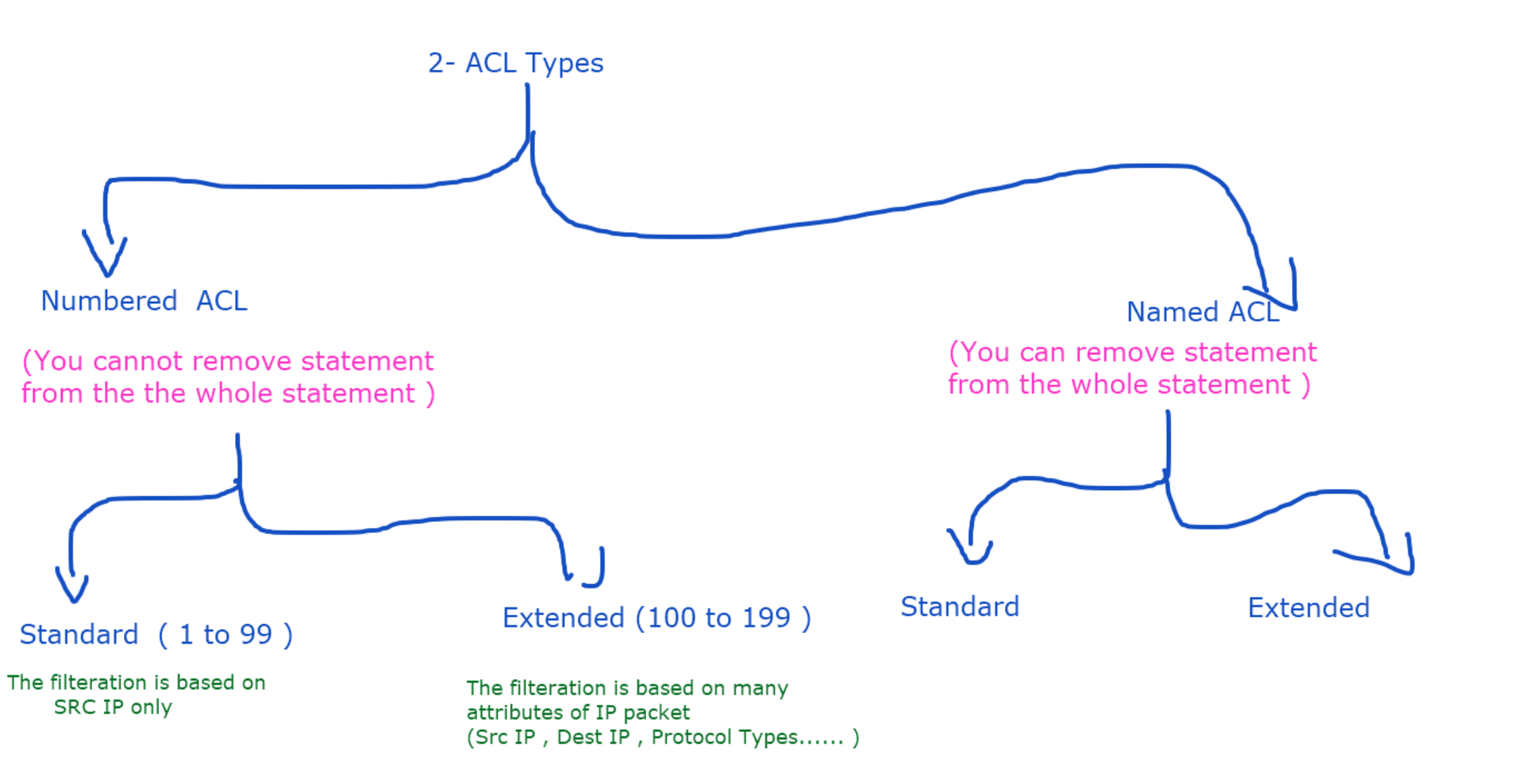
Security Through Cisco Router
Access Contril List

1-Def

It is a set of statement which grouped under specific numbered or named which is used to control IP traffic . each statement has one of two action (Allow or Deny)




2- ACL Types
Numbered ACL
(You cannot remove statement from the the whole statement)
Named ACL
(You can remove statement from the whole statement)
Standard (1 to 99)
Extended (100 to 199)
Standard
Extended
The filteration is based on SRC IP only
The filteration is based on many attributes of IP packet
(Src IP , Dest IP , Protocol Types.....)

3- ACL processing

1-Statements are processed fro up to down

2-One the matching is found , no other statements are processed

3-Any ACL must have at least one permit statements , otherwise the traffic will be dropped due to implicity deny any at the end of the list

4-Any new Statement must be added at the end of the list befor deny any
You cannot add statement between the whole statements by default.

5-The direction of ACL may be inbound or outbound

6-At the inbound ACL , The filteration is occued befor routing proceess but at outbound ACL , the filteration is occured after routing process

7-Max Number of ACL is one per interface per direction



Wildcard mask is inverted subnet mask . It is continous of zero's followed by continous of one's

Zero's means exact match

One's means donot care

192.168.33.1

0 . 0 . 0 .0



host 192.168.33.1

192.168.33.0

0 .0 .0 .255



192.168.33.0

0. 0. 0.255

0 . 0. 0. 0

255.255.255.255



ANY

1-Standard Numbered ACL

A)Properties

1-It has a numbered from 1 to 99

2-It is numbered and so you cannot remove the statement from the whole statements

3-It is stanadard and so the filteration is based on SRC IP only

4-It is created near to Destination as soon as possible

B)Configuration

Create Standard Numbered ACL Which must do the follwing
Deny all machines at 192.168.1.0 to access 192.168.13.0 except
192.168.1.1 which must be allowed . other all traffic must be permitted

ACL Creation

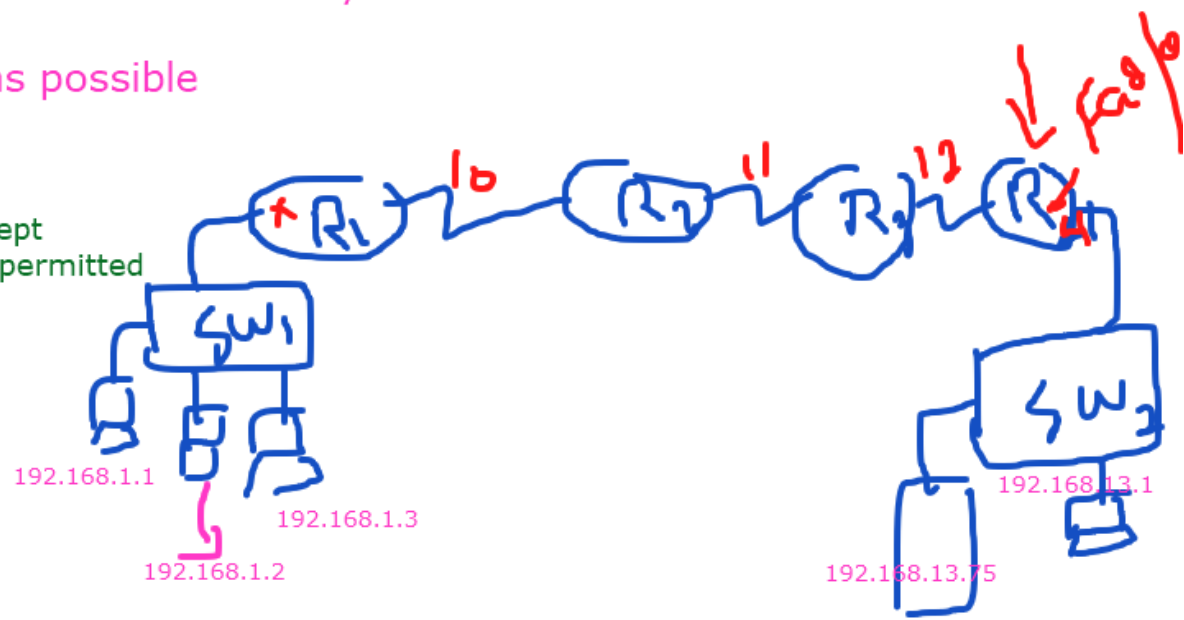
```
R4(config)#access-list 1  permit  host 192.168.1.1
R4(config)#access-list  1  deny  192.168.1.0 0.0.0.255
R4(config)#access-list  1 permit  Any
R4(config)#interface fa0/0
R4(config-if)#ip access-group 1 out
```

```
R4#show access-list
```

Tracing

1-from pc Tracert 192.168.13.1

2-from Router traceroute 192.168.13.1



2-Extended Numbered ACL

A)Properties

1-It has a number from 100 to 199

2-It is Numbered and so You cannot remove statement from the whole statement

3-Filteration is based on may attributes for IP packet (Src-IP , Dest-IP , Protocol type ....etc)

4-It is created near to Source as soon as possible

B)Configuration

TCP
UDP
ICMP
IP

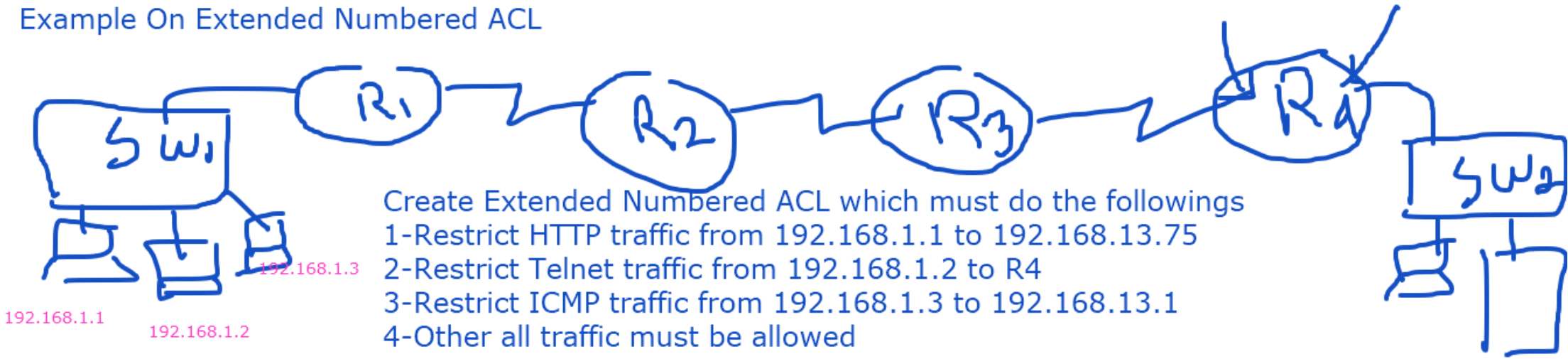
R(config)#Access-list 100 permit TCP/IP Src Dest Eq Port-Number

Restrict HTTP from 192.168.1.1 to 192.168.13.75

R1(config)#access-list 100 deny TCP 192.168.1.1 0.0.0.0 192.168.13.75 0.0.0.0 Eq 80



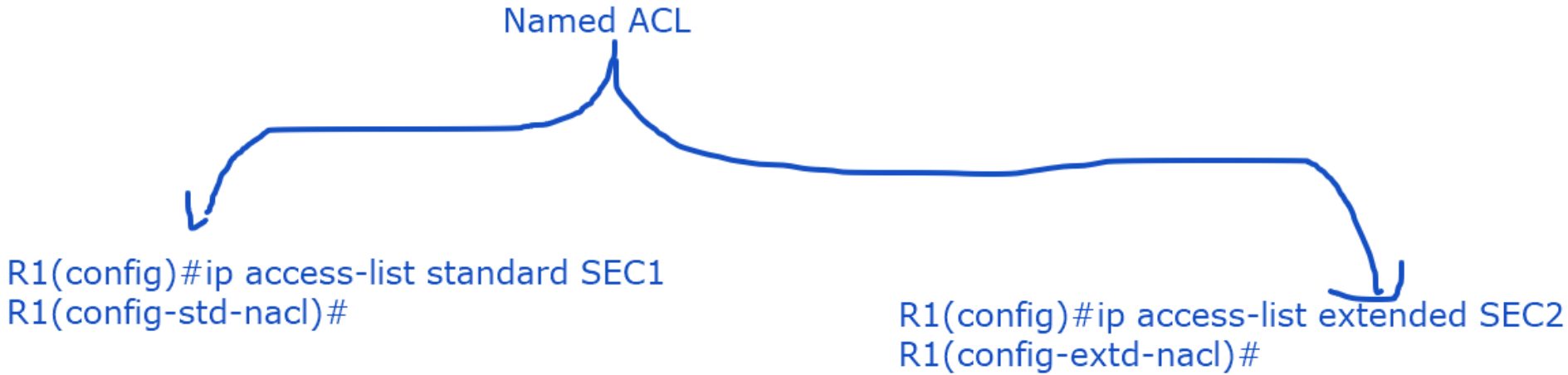
Example On Extended Numbered ACL



Create Extended Numbered ACL which must do the followings
1-Restrict HTTP traffic from 192.168.1.1 to 192.168.13.75
2-Restrict Telnet traffic from 192.168.1.2 to R4
3-Restrict ICMP traffic from 192.168.1.3 to 192.168.13.1
4-Other all traffic must be allowed

```
R1(config)#Access-list 100 deny TCP host 192.168.1.1 host 192.168.13.75  eq 80
R1(config)#access-list 100 deny TCP host 192.168.1.2 host 192.168.13.100 eq 23
R1(config)#access-list  100 deny TCP host 192.168.1.2 host 192.168.12.1 eq  23
R1(config)#access-list   100 deny ICMP host 192.168.1.3  host 192.168.13.1
R1(config)#access-list  100 permit  IP  Any Any

R1(config)#interface fa0/0
R1(config-if)#ip access-group 100  in
```


Named ACL
R1(config)#ip access-list standard SEC1
R1(config-std-nacl)#
R1(config)#ip access-list extended SEC2
R1(config-extd-nacl)#